

HNSS11 · Eye Pathology — Disease Chart

Vascular retinopathy · Retinal detachment · Optic nerve · Intraocular tumors · Orbit & proptosis | **STEP 1** tags mark rows/content beyond the lecture

HoyaDocs

Updated Oct 1, 2026 · Rev 1

Vascular Retinopathy & Retinal Neovascularization

DISEASE	WHO / RISK	PATHOGENESIS	CLINICAL & FUNDUS FINDINGS	PATHOLOGY (GROSS & MICRO)	KEY CLUES & DDX	PROGNOSIS & MANAGEMENT
Non-proliferative diabetic retinopathy <i>the leaky capillaries</i> Diabetes	Long-standing DM; #1 cause of blindness in adults (ages 24–63)	Endothelial / pericyte damage → microaneurysms , hemorrhage, leaking vessels → edema (macula especially prone) → lipid precipitates	Microaneurysms, flame (superficial) + dot-blot (deep) hemorrhages, hard exudates , cotton wool spots , macular edema (central acuity loss, periphery intact)	Microaneurysms on trypsin digest ; hard exudate = lipid; cotton wool spot = nerve-fiber-layer ischemia (blocked axoplasmic transport)	Same pattern in DM, HTN, vasculitis, radiation, leukemia → check glucose, BP, vasculitis/infection, radiation history	Macular edema = a main cause of permanent vision loss; anti-VEGF / laser STEP 1
Proliferative diabetic retinopathy <i>the vessels that grow the wrong way</i> Diabetes	Long-standing DM with CKD, MI; ↑ HbA1c (e.g., 9.6%)	Capillary closure → ischemia → VEGF → neovascularization that grows into the vitreous , NOT the retina	New vessels on disc/retina; vitreous hemorrhage (sudden dark vision): boat-shaped in young/type 1 (solid vitreous), diffuse haze in older patients	Fragile vessels bleed, leak, then scar and contract (like a burn scar) → pull the retina off the RPE	Acute loss = vitreous hemorrhage ; permanent loss = macular edema + traction RD	Traction retinal detachment = end stage; anti-VEGF, laser, vitrectomy STEP 1
Hypertensive retinopathy <i>the narrowed, nicked arterioles</i> Hypertension	Long-standing HTN; LVH, heart failure; normal IOP	Chronic pressure → arteriolosclerosis (hyaline) → narrowing, leakage, ischemia	Arteriolar narrowing , AV nicking , copper/silver wiring , flame hemorrhages , cotton wool spots, hard exudates	Hyaline arteriolosclerosis of retinal arterioles	Malignant HTN → disc swelling (optic nerve swelling) + flame hemorrhages	Control BP
Sickle cell retinopathy <i>the peripheral sea fan</i> Hemoglobinopathy	Young patient with pain crises ; normal glucose	Sickled RBCs occlude vessels, especially in the periphery (lower O ₂ tension promotes sickling) → ischemia → VEGF → neovascularization	Peripheral neovascularization (" sea fan "), vitreous hemorrhage	Neovascular fronds at the ischemic edge	Lecture vignette: young woman, severe abdominal pain + peripheral neovascularization + normal glucose → sickle cell . Don't reflexively call it diabetes	Can also cause traction RD
Central retinal artery occlusion <i>the eye stroke</i> Vascular occlusion	Older; atherosclerosis, carotid / cardiac emboli STEP 1	Arterial blockage → inner retinal ischemia	Sudden , painless vision loss	Pale, edematous retina; cherry-red spot at the fovea STEP 1	Another cause of sudden vision loss besides vitreous hemorrhage and RD	Treat as a stroke equivalent — urgent stroke workup
Central retinal vein occlusion <i>blood and thunder</i> Vascular occlusion STEP 1	Older; HTN, DM, glaucoma, hypercoagulable	Venous outflow blocked → hemorrhage, edema, ischemia → VEGF	Sudden painless loss; hemorrhages in all four quadrants , dilated tortuous veins	—	vs CRAO : pale retina + cherry-red spot	Risk of neovascular glaucoma; anti-VEGF

DISEASE	WHO / RISK	PATHOGENESIS	CLINICAL & FUNDUS FINDINGS	PATHOLOGY (GROSS & MICRO)	KEY CLUES & DDX	PROGNOSIS & MANAGEMENT
Retinopathy of prematurity <i>the oxygen-damaged newborn retina</i> Preterm STEP 1	Preterm infant on supplemental O ₂	High O ₂ halts vessel growth → later relative hypoxia → VEGF → neovascularization	Peripheral avascular retina; late leukocoria , RD	—	Leukocoria DDX: retinoblastoma, congenital cataract, ROP	Screening; laser / anti-VEGF

Retinal vocabulary & histology the lecture emphasized

TERM	WHAT IT IS	WHY IT MATTERS
Retinal layers (3 emphasized)	Photoreceptors (rods, cones) → bipolar cells → ganglion cells , whose axons run along the retinal surface and converge into the optic nerve	Retinal detachment = neurosensory retina (inner optic-cup layer) separating from the RPE (outer layer) — two embryologically adjacent layers
Intraretinal hemorrhage	Bleeding within the retina; its shape is set by the layer it's in	Superficial (nerve fiber layer) = flame / splinter ; deep = dot-blot (round)
Vitreous hemorrhage	Bleeding into the vitreous cavity (not the retina) → sudden dark/obscured vision	Young (e.g., type 1 DM): solid vitreous traps blood on the retina → boat-shaped bulge . Older : liquefied vitreous → diffuse haze
Microaneurysm	Tiny balloon-like capillary dilation; leaks fluid/lipoprotein, bleeds	Seen on trypsin digest (retina digested away to show the capillary network)
Hard exudate	Shiny lipid/lipoprotein left after capillaries leak (around microaneurysms)	Marks leakage
Soft exudate (cotton wool spot)	Nerve-fiber-layer ischemia blocks axoplasmic transport → white, fluffy swelling	Nonspecific (DM, HTN, vasculitis, radiation, leukemia); no significant vision loss itself
Macular edema	Leaky vessels → fluid in the macula → thickening	Severe central acuity loss even with an intact peripheral field
Neovascularization	Ischemia → VEGF → new vessels from the retinal surface into the vitreous	Fragile: bleed, leak, scar and contract

★ **USMLE high-yield:** Diabetic retinopathy: **acute** vision loss = **vitreous hemorrhage**; **permanent** = **macular edema + traction RD**. Neovascularization grows **into the vitreous**. Cotton wool spot = nonspecific microvascular ischemia (DM, HTN, vasculitis, radiation, leukemia). Hard exudate = lipid. Peripheral “sea fan” + normal glucose = sickle cell. **Retinal artery occlusion = stroke equivalent**. Workup of retinal hemorrhage/exudate: **glucose, BP, vasculitis/infection, head & neck radiation**.

Retinal Detachment — neurosensory retina separates from the RPE

DISEASE	WHO / RISK	PATHOGENESIS	CLINICAL & FUNDUS FINDINGS	PATHOLOGY (GROSS & MICRO)	KEY CLUES & DDX	PROGNOSIS & MANAGEMENT
Tractional retinal detachment <i>the scar that pulls</i> Traction	Proliferative DR (end stage), sickle cell	Fibrovascular scar from neovascularization contracts and pulls the retina off	Sudden painless field loss, “ shade/curtain pulled across” (field defect opposite the detached retina)	Retina off the RPE; photoreceptors lose choriocapillaris supply → degenerate; RPE proliferates	Diabetic + sudden loss of the upper field → inferior traction RD	Vitrectomy; early reattachment preserves vision STEP 1
Rhegmatogenous retinal detachment <i>the tear that lets fluid in</i> Tear	Myopia, trauma, cataract surgery, older adults (aging vitreous)	Retinal tear → vitreous fluid passes under the retina	Flashes + floaters → curtain-like field loss	Same separation plane (embryonic optic vesicle cavity)	Most common type STEP 1	Emergency, especially if the macula is still attached STEP 1
Exudative (serous) retinal detachment <i>the tumor that lifts</i> Exudate / tumor	Choroidal melanoma or other mass; malignant HTN; inflammation	Fluid leaks under an intact retina; a uveal mass lifts the overlying retina	Painless field loss (“shade”) in an adult with a uveal mass	Subretinal fluid over the mass	Lecture review Q: 61-year-old, “shade pulled across,” uveal mass → melanoma	Treat the cause

★ **USMLE high-yield: Three ways the retina comes off: Tear, Traction, Tumor (exudate).** Sudden painless monocular loss DDX: RD (curtain), CRAO (cherry-red spot), CRVO (blood and thunder), vitreous hemorrhage.

Optic Nerve Disorders — read the disc: cupped, pale or swollen (not reached in class; still on the LOs)

DISEASE	WHO / RISK	PATHOGENESIS	CLINICAL & FUNDUS FINDINGS	PATHOLOGY (GROSS & MICRO)	KEY CLUES & DDX	PROGNOSIS & MANAGEMENT
Glaucoma <i>the cupped disc</i> Cupping	↑ IOP; older; African descent STEP 1	Loss of ganglion cell axons with posterior bowing of the lamina cribrosa	Peripheral field loss first; acute angle-closure = painful red eye, fixed mid-dilated pupil STEP 1	Glaucomatous “cupping”: enlarged, excavated cup	Cup = chronic pressure	Lower IOP
Optic atrophy (non-glaucomatous) <i>the pale disc</i> Pallor	Ischemia (AION, giant cell arteritis), compression (pituitary adenoma, meningioma, orbital mass, Graves), optic neuritis/MS, chronic papilledema, trauma, toxins, LHON	Loss of axons and capillaries → gliosis	Progressive or sudden vision loss depending on cause; bitemporal hemianopia with pituitary STEP 1	Optic nerve pallor (flat, pale disc)	Pale = everything except glaucoma	Find and treat the cause
Papilledema <i>the disc swollen by brain pressure</i> ↑ ICP	Brain tumor, hemorrhage, IIH (obese young woman; vitamin A, tetracyclines) STEP 1	High CSF pressure in the optic nerve sheath blocks axoplasmic flow → axons swell at the disc	Bilateral disc swelling; headache worse lying down, vomiting; vision preserved early (enlarged blind spot)	Swollen disc, blurred margins	Papilledema is ONE cause of optic nerve swelling: the ↑ ICP subset	Chronic → optic atrophy; treat ICP (acetazolamide for IIH STEP 1)
Optic nerve swelling (not papilledema) <i>the swollen disc from another cause</i> Other swelling	Severe hypertension; infiltration by tumor or granuloma (leukemia, metastases, sarcoidosis); optic neuritis, ischemic optic neuropathy STEP 1	Hypertensive or infiltrative injury to the nerve head	Often unilateral; vision often reduced (optic neuritis: painful eye movement, color desaturation STEP 1)	Swollen disc ± infiltrating cells	All papilledema is optic nerve swelling, but not all swelling is papilledema	Treat underlying cause

★ **USMLE high-yield: Cupped** = glaucoma · **pale** = non-glaucomatous atrophy (ischemic, compressive...) · **swollen** = papilledema (↑ ICP, bilateral, vision spared early) or another cause (severe HTN, infiltration). “Papill-EDEMA = ICP-EDEMA.”

Intraocular Tumors

DISEASE	WHO / RISK	PATHOGENESIS	CLINICAL & FUNDUS FINDINGS	PATHOLOGY (GROSS & MICRO)	KEY CLUES & DDX	PROGNOSIS & MANAGEMENT
Choroidal (uveal) melanoma <i>the adult eye's own cancer</i> Malignant	Adults; most common primary intraocular malignancy in adults ; eye = 2nd most common primary melanoma site after skin	Melanocytes of the uvea (choroid > ciliary body > iris)	Pigmented (or not) dome/mushroom mass; exudative RD ("shade across vision")	Callender cytology: epithelioid (worse), spindle (more favorable), mixed (intermediate)	Prognosis: basal diameter (not height), scleral invasion , cytology , chromosome 3 . Color is NOT important	Spreads hematogenously to liver (no uveal lymphatics); GNAQ/GNA11, BAP1 STEP 1
Uveal nevus & Lisch nodules <i>the benign look-alikes</i> Benign	Common, especially in light-colored eyes ; Lisch nodules → neurofibromatosis	Benign melanocytic proliferation; Lisch nodule = variant of iris nevus	Flat, small, stable pigmented lesion (like a skin freckle/mole); Lisch nodules = raised , numerous small iris nodules	More benign cytology than melanoma	Lisch nodules + café-au-lait spots + neurofibromas → NF1 (chr 17) STEP 1	Observe
Metastasis to the uvea <i>the most common intraocular tumor overall</i> Malignant	Known cancer: lung and breast most common	Hematogenous spread to the highly vascular uvea (choroid)	Often amelanotic , multiple, bilateral; can look like amelanotic melanoma	Histology of the primary tumor	Resembles amelanotic melanoma ; intraocular biopsy is difficult → ultrasound distinguishes: metastasis hyperechoic , melanoma hypoechoic	Most common intraocular malignancy overall STEP 1
Leukemic / lymphomatous infiltration <i>the flat white choroid</i> Infiltrative	Known leukemia or lymphoma	Malignant cells infiltrate the uvea (choroid)	White, relatively flat choroidal lesions beneath the retinal vessels	Sheets of leukemic / lymphoid cells	DDx look-alikes: other infiltrative/inflammatory disease — syphilis, TB, sarcoidosis	Treat the systemic disease
Retinoblastoma <i>the child's white pupil</i> Malignant	Children; most common primary intraocular malignancy in children ; very lethal. A retinal/neural tumor (unlike choroidal melanoma)	Loss of both RB1 copies (13q14) STEP 1 ; hereditary = germline, bilateral, early	Leukocoria (white reflex; check red reflex at well-baby exams) and strabismus (needs dilated fundus exam); grows fast, invades optic nerve → brain	White, friable, calcified mass; Flexner-Wintersteiner rosettes, Homer-Wright rosettes, fleurettes	Prognosis: size + vitreous seeding, optic nerve invasion, extrascleral invasion, histology	Enucleation: clear optic nerve margin → local treatment; tumor at margin → intrathecal chemo . Fleurettes/F-W = favorable; survivors of hereditary RB → osteosarcoma STEP 1

Retinoblastoma rosettes

PATTERN	CENTER	SEEN IN	MEANING
Flexner-Wintersteiner rosette	Empty lumen	Retinoblastoma (± pineoblastoma)	Photoreceptor differentiation; characteristic of RB; better differentiated → favorable
Homer-Wright rosette	Fibrillary neuropil , no lumen	Retinoblastoma, neuroblastoma, medulloblastoma, PNET	Less differentiated ("island of blue cells in a sea of pink necrosis") → worse prognosis ; less specific
Fleurettes	"Bouquet of flowers"	Retinoblastoma	Recapitulate photoreceptors; most differentiated → good prognosis

★ **USMLE high-yield:** Adult intraocular mass → **choroidal melanoma** (primary) vs **metastasis** (lung, breast; most common overall). Child + leukocoria → **retinoblastoma**. Melanoma: **epithelioid = evil, spindle = safer**; color doesn't matter; spreads to **liver**. F-W rosette has a **window** (lumen); H-W has **hair** (neuropil); fleurettes are **favorable**.

Orbit & Proptosis (Exophthalmos)

DISEASE	WHO / RISK	PATHOGENESIS	CLINICAL & FUNDUS FINDINGS	PATHOLOGY (GROSS & MICRO)	KEY CLUES & DDX	PROGNOSIS & MANAGEMENT
Graves orbitopathy <i>the hyperthyroid bulging eyes</i> (Autoimmune)	Adults (often women); most common cause of proptosis	TSH-receptor antibodies activate orbital fibroblasts → GAG deposition STEP 1	Bilateral proptosis, diplopia (restricted movement), swelling, irritation; palpitations, sweating, weight loss; severe → optic nerve compression	Enlarged extraocular muscles (inferior + medial rectus most), tendons spared; lymphocytes among muscle fibers, GAGs, edema → fibrosis	Lecture vignette: 35-year-old man, bilateral diplopia/swelling + hyperthyroid symptoms	Treat thyroid; steroids, decompression STEP 1
Idiopathic orbital inflammation (orbital pseudotumor) <i>the painful mystery mass</i> (Inflammatory)	Adults; no thyroid or systemic disease	Non-specific, non-infectious orbital inflammation	Painful , often unilateral proptosis	Polymorphous inflammation + fibrosis	Diagnosis of exclusion	Steroid-responsive
Infectious orbital cellulitis <i>the infected orbit</i> (Infectious)	Often spread from (ethmoid) sinusitis	Bacterial infection of orbital tissue	Fever, pain on eye movement, ↓ vision, proptosis	Acute inflammation (neutrophils) and pus	vs preseptal cellulitis: normal eye movements and vision STEP 1	Dangerous: orbital veins drain to the cavernous sinus → intracranial spread
Orbital lymphoma <i>the adult's orbital cancer</i> (Malignant) NOT COVERED IN CLASS	Older adults	Usually B-cell (MALT) lymphoma STEP 1	Painless, slowly progressive proptosis; "salmon patch" STEP 1	Sheets of monotonous lymphocytes	Most common primary orbital malignancy in adults	Radiation / chemotherapy STEP 1
Rhabdomyosarcoma <i>the child's rapid proptosis</i> (Malignant) NOT COVERED IN CLASS	Children	Arises from orbital soft tissue , NOT the extraocular muscles	Rapid (weeks), unilateral proptosis, eyelid swelling	Ranges from primitive small round blue cells to cells with skeletal-muscle cross-striations ; desmin, myogenin STEP 1	Most common malignant primary orbital tumor in children	Chemo + radiation STEP 1
Systemic inflammation / vasculitis <i>the granulomatous orbit</i> (Systemic)	Sarcoidosis, GPA	Granulomatous inflammation / necrotizing vasculitis	Proptosis with systemic disease (hilar nodes; sinusitis + glomerulonephritis)	Non-caseating granulomas (sarcoid); necrotizing granulomatous vasculitis (GPA)	c-ANCA in GPA STEP 1	Treat systemic disease

Orbital tumors by age — don't memorize all; histology is often needed for diagnosis

AGE	BENIGN	MALIGNANT	METASTATIC
Children	Lymphangioma, capillary hemangioma, dermoid, optic nerve glioma (astrocytoma)	Rhabdomyosarcoma , histiocytosis X	Neuroblastoma, leukemia
Adults	Cavernous hemangioma, mucocoele, optic nerve meningioma, lacrimal gland tumors	Lymphoma , lacrimal gland tumors (similar to salivary gland pathology), soft tissue tumors, neurofibroma, schwannoma	Breast, lung; invasion from sinuses, nasopharynx, cranial cavity

★ **USMLE high-yield:** Orbital disease presents with **proptosis, diplopia, optic nerve compression**. Proptosis DDX: **Graves** (bilateral, hyperthyroid, fat muscle bellies) → **pseudotumor** (painful, steroid-responsive) → **cellulitis** (neutrophils, pus, fever) → **tumor** (adult = **lymphoma**; child = **rhabdomyosarcoma**) → **sarcoid/GPA**.